COP 3710 - Course Project Description

with Service Learning Component & Digital Badge

FORMAT

- Group of three (one team may be a group of two)
- Report submission covering all aspects outlined below
- 10-15 minute presentation
- **Service Learning component with NGO partnership (Required)**
- **Qualifies for FGCU Data Literacy Transferable Skills Digital Badge**

NOTES

- Students have a choice to select their group; otherwise, students will be randomly assigned. The deadline for group selection is January 21, 2026.
- If a student drops the course mid-semester, the remaining students will continue with the project and select a new group lead.
- If a student in the group does not perform, other students can report to the instructor. Three peer evaluations are included to report such incidents immediately rather than waiting until the end.
- Except for peer evaluation, the group lead must do all submissions.
- Each group's project must be unique and distinct from other groups. In the event of similarity between projects, both groups will be required to revise and resubmit their work.
- **All groups must partner with a different NGO/community partner, or different projects for the same NGO/community partner, to ensure diverse learning experiences.**

DATA LITERACY DIGITAL BADGE OPPORTUNITY

This course project has been designed to align with FGCU's Data Literacy Transferable Skills Digital Badge. Upon successful completion of this project, students will have demonstrated competencies in:

- Collecting and organizing data from real-world sources
- Designing and implementing database systems
- Analyzing data to generate meaningful insights
- Communicating data-driven recommendations to stakeholders
- Using data to solve organizational problems

Digital Badge Benefits:

- Enhance your resume and LinkedIn profile with verifiable skills
- Demonstrate career readiness to potential employers
- Stand out in competitive job markets
- No additional cost or courses required

Your final project report will serve as the artifact for badge submission. More information about applying for the Data Literacy badge will be provided during the semester.

Learn more: https://www.fgcu.edu/digitalbadges/#transferableskills

PROJECT OUTLINE (Total 23 points)

1. SERVICE LEARNING PARTNERSHIP IDENTIFICATION AND INITIAL MEETING (3 points)

Project Objective: Identify and establish a partnership with a non-profit or government agency through FGCU's Service Learning program to address their real-world data management needs.

Requirements:

a) Partner Selection (1 point)

- Use the FGCU Community Partner Database (https://servicelearning.fgcu.edu/CommunityPartnerDatabase/agencieslist) to identify potential NGO partners
  Here is the link to the potential partnerlist
  https://docs.google.com/spreadsheets/d/1JbRFTsUCPJI0gzzJIx4Mk6zNY6-_ZyM5spn1uT0TbVA/edit?usp=sharing
  If you can't find a partner, reach out to the Service Learning Office Contact: Jamie Wilson (jawilson@fgcu.edu), to help you find a community partner
- Select an organization whose mission aligns with your team's interests
- Contact the organization and confirm their interest in partnering for a database project
- Submit partner confirmation form including: organization name, contact person, email, phone, and brief description of their mission

b) Needs Assessment Meeting (1 point)

- Schedule and conduct an initial meeting with the NGO representative (virtual or in-person)
- Document the organization's current data challenges and needs
- Identify what data they collect, how they currently manage it, and what problems they face
- Understand their reporting requirements, stakeholder needs, and future goals
- Submit meeting notes including: date, attendees, key discussion points, and identified data needs

c) Eagle Service Network Registration (1 point)

- Create a Service Learning form in Eagle Service Network (https://fgcu-servicelearning-csm.symplicity.com/students/)
- Include project details and community partner information
- Begin tracking all hours spent on this project (meetings, research, development, documentation)
- Submit screenshot of approved ESN form

Best Practices:

- Come prepared to the meeting with questions about their data and processes
- Be professional and respectful of the partner's time
- Maintain regular communication throughout the semester
- Remember that you are representing FGCU

Due Date: Feburary 2, 2026

2. INTRODUCTION AND PROJECT PROPOSAL (2 points)

Requirements:

- **Project Objective:** Clearly state the purpose and expected outcomes of the database project based on the NGO partner's identified needs. Include how this database will benefit the organization and their constituents.
- **Community Partner Overview:** Provide background on the NGO including their mission, services, target population, and current data management challenges.
- **Scope of the Project:** Define the boundaries and scale of the project, including what aspects will be covered and what will be out of scope. This should be agreed upon with the community partner.
- **Dataset Description:** Explain the types of data you will manage (based on partner needs). Try to use a variety of data types that reflect real organizational data.
- **Impact Statement:** Describe how this database system will improve the NGO's operations, decision-making, or service delivery.
- **Partner Approval:** Include a statement or email confirmation from the community partner approving the proposed project scope.

Due Date: February 2, 2026 (Week 4)

3. TABLES CREATION (2 points)

Requirements:

- Write SQL scripts to create tables in PostgreSQL. Include primary and foreign key constraints, check constraints, and any other relevant constraints based on business rules.
- Provide well-commented SQL scripts explaining the purpose of each table and constraint in the context of the NGO's needs.
- Ensure table structures accommodate the actual data the organization will need to store.
- Based on your understanding of the partner's needs, design tables that will effectively organize their data.

Submission: Provide SQL file for table creation.
Due Date: Feb 25, 2026 (Week 8)

4. DATA POPULATION (2 points)

Requirements:

- Populate the tables with realistic sample data that reflects the type of information the NGO will manage. Include sufficient data to demonstrate all aspects of the database (minimum 15-20 records per major table).
- Work with your community partner to understand realistic data scenarios, patterns, and edge cases.
- Discuss and implement methods to maintain data integrity and consistency.
- If using any real organizational data, ensure it is properly anonymized and you have written permission.

Submission: Provide SQL file for data insertion.
Due Date: Feb 25, 2026 (Week 8)

5. MID-PROJECT PARTNER CHECK-IN (1 point)

Requirements:

- Schedule a second meeting with your community partner to review progress

- Present your table structures and sample data
- Gather feedback and make necessary adjustments
- Discuss query requirements and what questions they need answered from the data
- Update project timeline and deliverables as needed

Submission: Meeting notes and updated project plan based on partner feedback.
Due Date: March 23, 2026 (Week 11)

6. SQL QUERIES (5 points)

Requirements:

- In collaboration with your community partner, develop a set of TEN different questions they would want to answer from their data.
- Include a mix of query types (e.g., selection, projection, joins, aggregations, subqueries, window functions).
- Each query should address a real business question or reporting need identified by the NGO.
- Provide explanations for each query, detailing its purpose, functionality, and how it benefits the organization.
- For each query, include sample results and explain what insights they provide to the organization.

Submission: Provide separate SQL files for your queries, mentioning query questions as comments and actual SQL queries. Include a document showing sample results and business insights for each query.
Due Date: March 23, 2026 (Week 11)

7. ENTITY-RELATIONSHIP (ER) DIAGRAM (2 points)

Requirements:

- Create an ER diagram representing the conceptual design based on the NGO's data requirements. Include entities, relationships, and cardinalities.
- Clearly indicate primary keys and explain the nature of relationships (one-to-many, many-to-many, etc.).
- Document the ER diagram based on your implemented tables and queries - this formalizes the design you've been working with.
- Share the ER diagram with your community partner and incorporate their feedback.
- Include a brief explanation of how each entity and relationship serves the organization's needs.

Submission: ER diagram (PDF/image) with documentation and partner feedback summary.
Due Date: April 8, 2026 (Week 14)

8. RELATIONAL MODEL (2 points)

Requirements:

- Convert the ER diagram into a relational schema, ensuring it aligns with your implemented tables and partner requirements.
- Apply normalization principles (up to 3NF or beyond) and define constraints (primary keys, foreign keys, check constraints, etc.).
- Create a comprehensive data dictionary documenting all tables, attributes, data types, constraints, and business rules.
- This should document and formalize the database structure you have already implemented.

Submission: Relational schema document with comprehensive data dictionary (PDF).
Due Date: April 8, 2026 (Week 14)

9. FINAL PARTNER PRESENTATION AND HANDOFF

Requirements:

- Schedule a final meeting with your community partner
- Present the complete database system and demonstrate its functionality
- Walk through each query and explain how to interpret results
- Provide all documentation, SQL scripts, and user guides
- Train the partner (if applicable) on basic database operations
- Discuss potential future enhancements or next steps
- Obtain final feedback and approval from the community partner

Deliverables to Partner:

- Complete database backup file
- All SQL scripts (table creation, data population, queries)
- ER diagram and relational schema documentation
- User guide or instructions for accessing/using the database
- Data dictionary
- Service hour log

Submission: Final presentation slides used with partner, partner feedback form/email confirming project completion, and reflection on partner's response.
Due Date: April 20, 2026 (Week 15)

10. PROJECT REPORT

(Mandatory for grading - Missing report will result in zero points for all sections)
Requirements:

- Comprehensive coverage of all project aspects with clear labeling and explanation
- Service Learning Integration section covering: community partner background, identified data needs, collaboration process, project impact, and total service hours
- Technical documentation including ER diagram, relational model, data dictionary, SQL scripts, and query results
- Comprehensive reflection on challenges, lessons learned, real-world application, personal growth, and career readiness connections
- Identification of relevant UN Sustainable Development Goals your project supports
- Eagle Service Network documentation (screenshots of approved form, verified hours, and partner evaluation)
- **This report will serve as your artifact for the Data Literacy Transferable Skills Digital Badge application**

Due Date: April 20, 2026 (Week 15)

11. CLASS PRESENTATION (2 points)

Requirements:

- Present community partner introduction, their mission, and the data challenges they faced
- Explain your database implementation approach and design decisions
- Demonstrate SQL queries, results, and organizational value with real examples
- Present ER diagram and relational model showing how they align with partner needs

- Share service learning experience including project benefits, skills developed, challenges, and career connections
- Include live demonstration if possible
- Conclude with key takeaways from technical and service learning aspects

Note: Presentation is mandatory for all teams; missing the presentation will result in zero points for the entire project.

Presentation Format: 10-15 minutes with Q&A
Due Date: April 22 and 27, 2026 (Week 16)

12. PEER EVALUATIONS

(Mandatory for grading - Missing evaluations will result in zero points for everyone)
Requirements:

- Three peer evaluation submissions to maintain checks and balances on group performance
- If a student is found not participating in any peer evaluation, that student will receive zero points for the entire project
- Other group members will be graded based on their project performance

Due Dates:

1. Evaluation 1 - February 4, 2026
2. Evaluation 2 - March 23, 2026
3. Evaluation 3 - April 20, 2026

13. EAGLE SERVICE NETWORK HOUR TRACKING (Mandatory)

Requirements:

- All group members must track their individual hours in Eagle Service Network
- Hours should include meetings, research, development, documentation, presentations, and all project activities
- Community partner must verify hours in Eagle Service Network
- Final verification must be completed by April 20, 2026
- Submit screenshots of approved ESN forms and verified hours as part of final report - *Mandatory for grading - Missing screenshot will result in zero points*

Note: These service learning hours will count toward your FGCU graduation requirement of 80 hours.

14 Digital Badge (2 Point)

The Digital Badge Portfolio is your opportunity to showcase your learning and growth by connecting your work (artifacts) to key competencies. Through thoughtful reflection, you will demonstrate how your experiences align with the program's learning outcomes.
https://www.fgcu.edu/digitalbadges/transferable
Note: You can use this assignment to aim for Data Literacy Digital Badge

Artifacts for Digital Badge

Include a minimum of one artifact per competency. An artifact can be a project, paper, presentation, video, or any other work that demonstrates your skills and knowledge. You can submit this project as an artifact for the Data Literacy Badge.

https://fgcu.formstack.com/forms/dataliteracy
Submit a screenshot for badge submission points in your report

ADDITIONAL SERVICE LEARNING REQUIREMENTS

Communication with Community Partners:

- Maintain professional and timely communication throughout the semester
- Respond to partner emails within 24-48 hours
- Schedule meetings at times convenient for the partner
- Send meeting agendas in advance and follow up with summary notes
- Keep your instructor informed of any communication challenges

Ethics and Professionalism:

- Treat all organizational data with confidentiality
- Obtain written permission before using any real data
- Anonymize any data used in presentations or reports
- Be reliable and meet all deadlines with the partner
- Be respectful of the organization's time and resources

Reflection and Learning:

- Throughout the project, consider how database concepts apply to real organizational needs
- Reflect on the social impact of your work
- Consider how data management can improve nonprofit effectiveness
- Think about career applications of this experience

RESOURCES

- FGCU Service Learning Faculty Resources: https://www.fgcu.edu/studentlife/servicelearning/facultyresources
- Community Partner Database: https://servicelearning.fgcu.edu/CommunityPartnerDatabase/agencieslist
- Eagle Service Network: https://fgcu-servicelearning-csm.symplicity.com/students/
- Service Learning Office Contact: Jamie Wilson (jawilson@fgcu.edu), Coordinator of Faculty Development and Partnerships
- FGCU Digital Badges - Transferable Skills: https://www.fgcu.edu/digitalbadges/#transferableskills
- Data Literacy Badge Application: https://fgcu.formstack.com/forms/dataliteracy

SUGGESTED PROJECT IDEAS WITH SERVICE LEARNING CONTEXT

1. Nonprofit Donor Management System
Partner Type: Local charity, foundation, or arts organization
Features: Donor tracking, donation history, campaign management, receipts, donor communication, acknowledgments

2. Food Bank Inventory and Distribution System
Partner Type: Food pantry, soup kitchen, or food rescue organization

Features: Inventory tracking, client intake, distribution records, volunteer management, reporting for grants

3. Animal Shelter Management System
Partner Type: Animal rescue, shelter, or wildlife rehabilitation center
Features: Animal intake/adoption, medical records, foster care tracking, volunteer coordination, donation management

4. Community Health Services Database
Partner Type: Community health center, mental health services, or wellness programs
Features: Client services, appointments, program enrollment, outcome tracking, referral management

5. Environmental Conservation Data System
Partner Type: Environmental nonprofit, conservation group, or sustainability organization
Features: Project tracking, volunteer activities, wildlife/habitat monitoring, educational program management, grant reporting

6. Youth Program Management System
Partner Type: After-school programs, youth sports, mentoring programs
Features: Participant enrollment, attendance tracking, parent communication, program activities, outcome measurements

7. Volunteer Coordination System
Partner Type: Volunteer center or large nonprofit
Features: Volunteer profiles, skills tracking, opportunity matching, hours logging, event coordination, impact reporting

8. Community Arts and Events Database
Partner Type: Community theater, arts council, or cultural center
Features: Event management, participant registration, equipment/venue scheduling, ticketing, donor/sponsor tracking

9. Educational Tutoring Program System
Partner Type: Literacy program, tutoring center, or homework help organization
Features: Student enrollment, tutor matching, session scheduling, progress tracking, resource management

10. Homeless Services Coordination System
Partner Type: Homeless shelter, transitional housing program, or outreach services
Features: Client intake, case management, service referrals, bed tracking, resource allocation, outcome reporting